Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: River Oak Dairy #2

Physical address of dairy:

9161 Warnerville RD	Oakdale	Stanislaus	95361
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1985

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X064-X030-X006-XXXX

B. OPERATORS

Koolhaas, Tom

Operator name: Koolhaas, Tom

Telephone no.: Landline: ; Cellular: (209) 324-1241

P.O. Box 294	Farmington	CA	95230
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Sikkema, Ralph

Legal owner name: Sikkema, Ralph

Telephone no.: Landline: (209) 848-9132; Cellular:

11409 Scarlett Oak RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	300	600	250	0
Number under roof	1,175	175	0	0	0	0
Maximum number	1,175	175	300	600	250	0
Average number	1,175	175	300	600	250	0
Avg live weight (lbs)	1,300	1,350	900	650		

Predominant milk cow breed: Other

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 42,391.25 *tons per reporting period*

Total nitrogen from manure: 527,800.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 369,460.07 *lbs per reporting period*

Total phosphorus from manure: 86,950.12 *lbs per reporting period*

Total potassium from manure: 226,567.29 *lbs per reporting period*

Total salt from manure: 593,490.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,615,000 *gallons*

Total nitrogen generated: 56,600.34 *lbs*

Total phosphorus generated: 7,386.79 *lbs*

Total potassium generated: 50,942.72 *lbs*

Total salt generated: 223,567.56 *lbs*

	8,615,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,615,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal OID	Ground water
IW 1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/15/2023	Corral solids	1,777.50 *ton*	Dry-weight	56.9		27,400.00	8,790.00	26,800.00		34.40
11/02/2023	Corral solids	2,615.00 *ton*	Dry-weight	41.5		25,500.00	13,400.00	17,600.00		54.40

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	120,000.94	54,466.05	94,911.17	2,191,473.72
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	120,000.94	54,466.05	94,911.17	2,191,473.72

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	10	10	1	process wastewater	X064-X029-X013-XXXX
2	10	10	1	process wastewater	X064-X029-X013-XXXX
3	10	10	1	process wastewater	X064-X029-X013-XXXX
East	40	40	2	process wastewater	X064-X030-X006-XXXX
West	45	45	2	process wastewater	X064-X030-X006-XXXX
Totals for areas that were used for application	115	115	7		
Totals for areas that were not used for application					
Land application area totals	115	115	7		

B. CROPS AND HARVESTS

1

Field name: 1

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	217.00 *ton*	As-is		63.4	4,470.00	810.00	5,540.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.70	194.00	35.15	240.44	1,493.13

2

Field name: 2

2

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	217.00 *ton*	As-is		63.4	4,470.00	810.00	5,540.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.70	194.00	35.15	240.44	1,493.13

3

Field name: 3

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	217.00 *ton*	As-is		63.4	4,470.00	810.00	5,540.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.70	194.00	35.15	240.44	1,493.13

East

Field name: East

East

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	866.09 *ton*	As-is		60.2	4,430.00	790.00	5,680.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.65	191.84	34.21	245.97	1,396.05

06/14/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 40 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	832.00 *ton*	As-is		65.4	6,090.00	1,050.00	13,300.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	240.00	0.00
Total actual harvest content	20.80	253.34	43.68	553.28	1,971.92

West

Field name: West

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	976.50 *ton*	As-is		57.0	5,770.00	940.00	6,170.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.70	250.42	40.80	267.78	1,604.93

West

06/14/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 45 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	1,119.00 *ton*	As-is		71.0	4,840.00	1,110.00	8,300.00		17.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	240.00	0.00
Total actual harvest content	24.87	240.71	55.20	412.79	2,581.66

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/28/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	212.30	51.32	260.36	916.28	300,000.00 *gal*
Application event totals		212.30	51.32	260.36	916.28	

2 - 10/28/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	212.30	51.32	260.36	916.28	300,000.00 *gal*
Application event totals		212.30	51.32	260.36	916.28	

3 - 10/28/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	212.30	51.32	260.36	916.28	300,000.00 *gal*
Application event totals		212.30	51.32	260.36	916.28	

East - 10/28/2022: Oats, silage-soft dough

Field name: East

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	106.15	25.66	130.18	458.14	600,000.00 *gal*
Application event totals		106.15	25.66	130.18	458.14	

03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.52	12.84	93.32	525.74	900,000.00 *gal*
Application event totals		62.52	12.84	93.32	525.74	

04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.38	13.33	21.71	175.56	765,000.00 *gal*
Application event totals		85.38	13.33	21.71	175.56	

East - 06/14/2023: Sudangrass, silage

Field name: East

Crop: Sudangrass, silage

Plant date: 06/14/2023

East - 06/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	207.62	5.43	42.46	136.19	320,000.00 *gal*
Application event totals		207.62	5.43	42.46	136.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	4,800,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	4,500,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.95	0.00	0.00	48.82	4,500,000.00 *gal*
Application event totals		0.95	0.00	0.00	48.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.99	11.19	224.56	1,027.81	1,035,000.00 *gal*
Canal OID	Ground water	0.70	0.00	0.00	35.80	3,300,000.00 *gal*
Application event totals		130.68	11.19	224.56	1,063.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.92	0.00	0.00	47.19	4,350,000.00 *gal*
Application event totals		0.92	0.00	0.00	47.19	

East - 06/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.92	0.00	0.00	47.19	4,350,000.00 *gal*
Application event totals		0.92	0.00	0.00	47.19	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.88	0.00	0.00	45.56	4,200,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

West - 10/28/2022: Oats, silage-soft dough

Field name: West

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	106.15	25.66	130.18	458.14	675,000.00 *gal*
Application event totals		106.15	25.66	130.18	458.14	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.69	13.70	99.54	560.78	1,080,000.00 *gal*
Application event totals		66.69	13.70	99.54	560.78	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.29	13.94	22.70	183.59	900,000.00 *gal*
Application event totals		89.29	13.94	22.70	183.59	

West - 06/14/2023: Sudangrass, silage

Field name: West

Crop: Sudangrass, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	207.62	5.43	42.46	136.19	360,000.00 *gal*
Application event totals		207.62	5.43	42.46	136.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	5,400,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.96	0.00	0.00	49.18	5,100,000.00 *gal*
Application event totals		0.96	0.00	0.00	49.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.93	0.00	0.00	47.73	4,950,000.00 *gal*
Application event totals		0.93	0.00	0.00	47.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.57	10.37	208.29	953.33	1,080,000.00 *gal*
Canal OID	Ground water	0.67	0.00	0.00	34.72	3,600,000.00 *gal*
Application event totals		121.24	10.37	208.29	988.05	

West - 06/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.93	0.00	0.00	47.73	4,950,000.00 *gal*
Application event totals		0.93	0.00	0.00	47.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.90	0.00	0.00	46.29	4,800,000.00 *gal*
Application event totals		0.90	0.00	0.00	46.29	

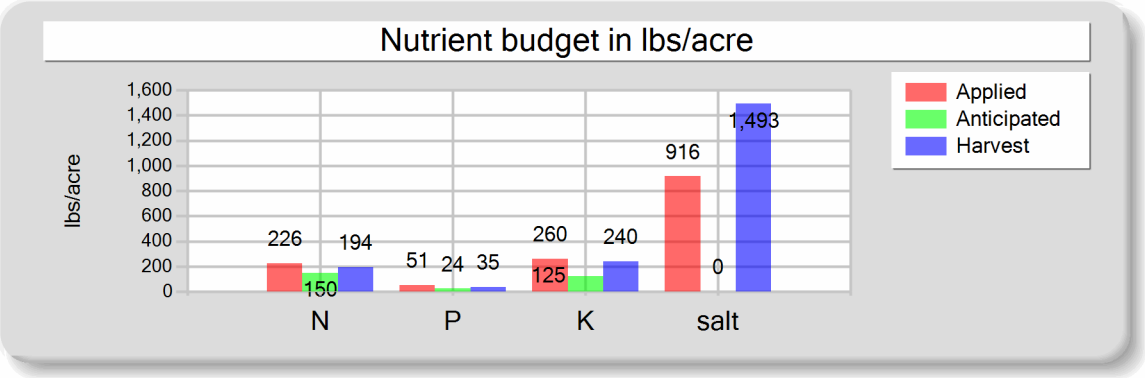
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.90	0.00	0.00	46.29	4,800,000.00 *gal*
Application event totals		0.90	0.00	0.00	46.29	

B. NUTRIENT BUDGET

1 - 10/28/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.30	51.32	260.36	916.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	226.30	51.32	260.36	916.28
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	194.00	35.15	240.44	1,493.13
Nutrient balance	32.30	16.17	19.93	-576.85
Applied to removed ratio	1.17	1.46	1.08	0.61

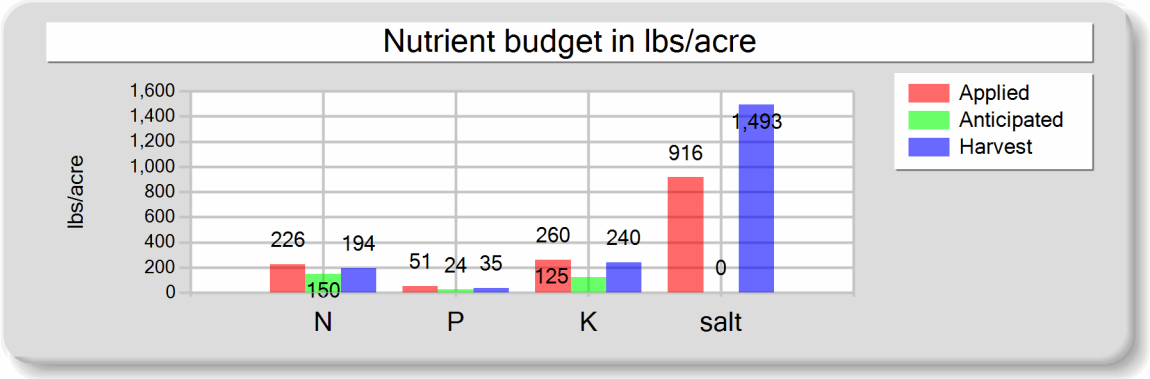
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
300,000.00 *gallons*
11.05 *acre-inches*
1.10 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 10/28/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.30	51.32	260.36	916.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	226.30	51.32	260.36	916.28
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	194.00	35.15	240.44	1,493.13
Nutrient balance	32.30	16.17	19.93	-576.85
Applied to removed ratio	1.17	1.46	1.08	0.61

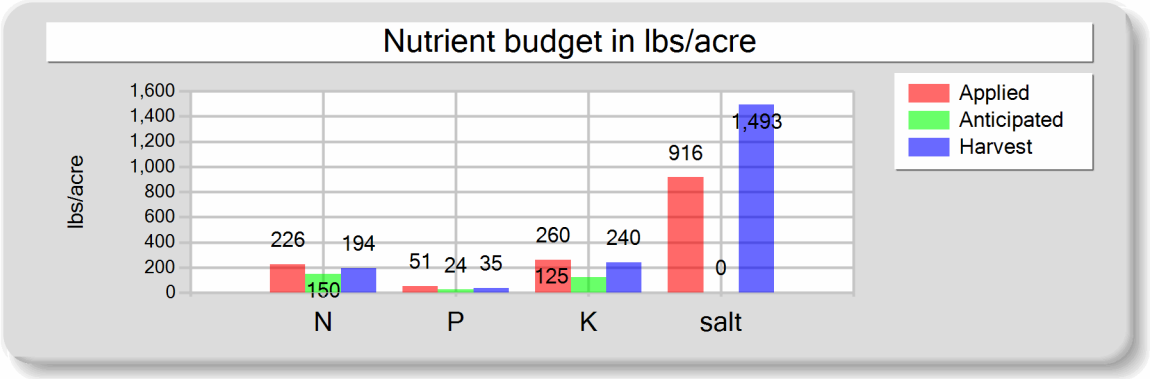
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
300,000.00 *gallons*
11.05 *acre-inches*
1.10 *inches/acre*

Total harvests for the crop
1 *harvests*

3 - 10/28/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.30	51.32	260.36	916.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	226.30	51.32	260.36	916.28
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	194.00	35.15	240.44	1,493.13
Nutrient balance	32.30	16.17	19.93	-576.85
Applied to removed ratio	1.17	1.46	1.08	0.61

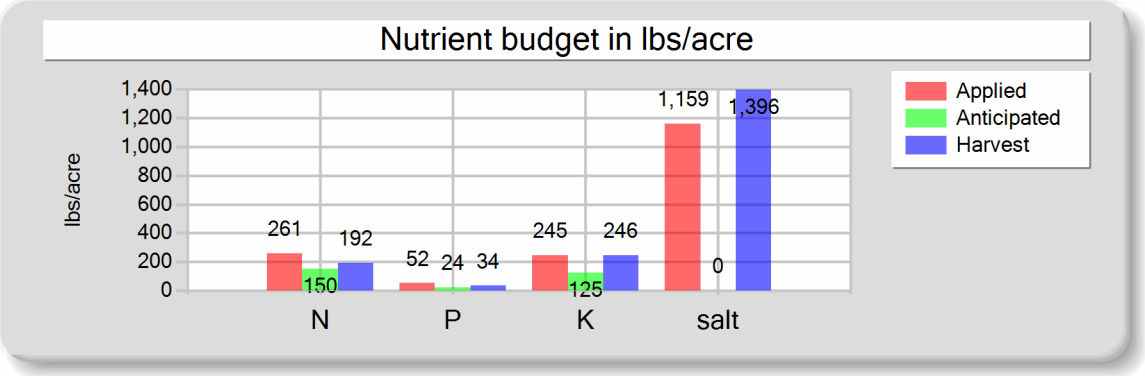
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
300,000.00 *gallons*
11.05 *acre-inches*
1.10 *inches/acre*

Total harvests for the crop
1 *harvests*

East - 10/28/2022: Oats, silage-soft dough

Field name: East | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	254.06	51.83	245.21	1,159.43
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.06	51.83	245.21	1,159.43
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	191.84	34.21	245.97	1,396.05
Nutrient balance	69.22	17.62	-0.76	-236.62
Applied to removed ratio	1.36	1.52	1.00	0.83

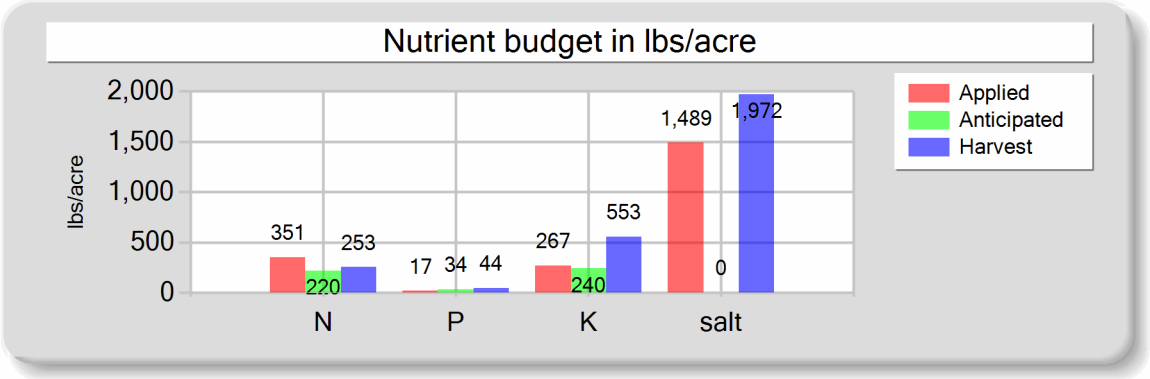
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,265,000.00	*gallons*
83.41	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

East - 06/14/2023: Sudangrass, silage

Field name: East | Crop: Sudangrass, silage | Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.61	16.61	267.02	1,164.00
Fresh water	6.32	0.00	0.00	325.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.93	16.61	267.02	1,489.46
Anticipated crop nutrient removal	220.00	34.00	240.00	0.00
Actual crop nutrient removal	253.34	43.68	553.28	1,971.92
Nutrient balance	97.59	-27.07	-286.26	-482.47
Applied to removed ratio	1.39	0.38	0.48	0.76

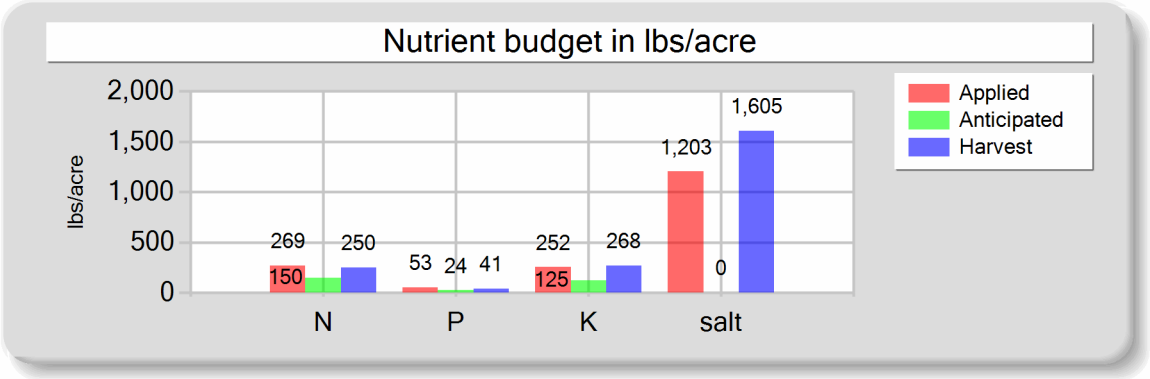
Fresh water applied	
30,000,000.00	*gallons*
1,104.80	*acre-inches*
27.62	*inches/acre*

Process wastewater applied	
1,355,000.00	*gallons*
49.90	*acre-inches*
1.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

West - 10/28/2022: Oats, silage-soft dough

Field name: West | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.13	53.30	252.42	1,202.51
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.13	53.30	252.42	1,202.51
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	250.42	40.80	267.78	1,604.93
Nutrient balance	18.72	12.50	-15.36	-402.42
Applied to removed ratio	1.07	1.31	0.94	0.75

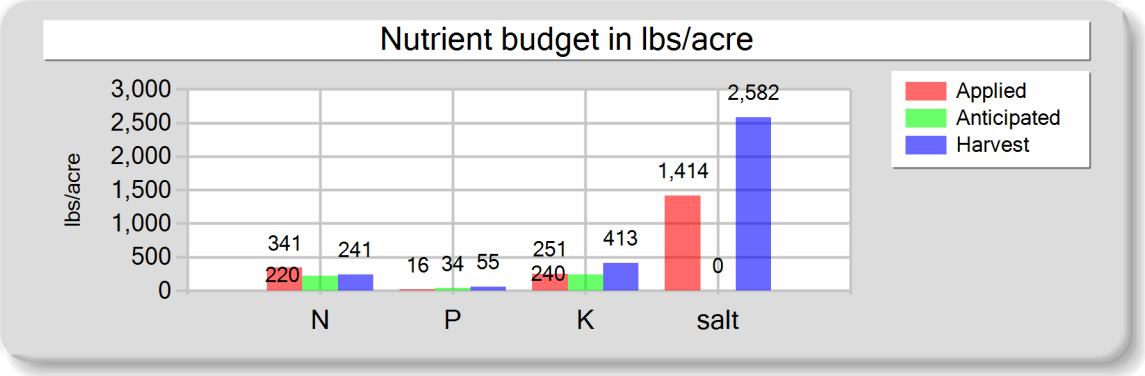
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,655,000.00	*gallons*
97.77	*acre-inches*
2.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

West - 06/14/2023: Sudangrass, silage

Field name: West | Crop: Sudangrass, silage | Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	328.19	15.80	250.75	1,089.52
Fresh water	6.29	0.00	0.00	324.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	341.49	15.80	250.75	1,413.53
Anticipated crop nutrient removal	220.00	34.00	240.00	0.00
Actual crop nutrient removal	240.71	55.20	412.79	2,581.66
Nutrient balance	100.78	-39.40	-162.04	-1,168.13
Applied to removed ratio	1.42	0.29	0.61	0.55

Fresh water applied	
33,600,000.00	*gallons*
1,237.37	*acre-inches*
27.50	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	5,630.00	17,400.00							65.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,400.00	8,790.00	26,800.00							34.40
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,500.00	13,400.00	17,600.00							54.40
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 10/17/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	848.00	521.00	0.00	0.00	205.00	1,040.00								7,390.00	3,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	333.00	168.00	0.00	0.00	68.40	497.00								4,040.00	2,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	535.00	168.00	0.00	0.00	92.70	626.00								4,200.00	3,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 06/29/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3,110.00	476.00	0.00	0.00	81.30	636.00								6,830.00	2,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	602.00	375.00	0.00	0.00	51.80	1,040.00								7,280.00	4,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	728.00	190.00	0.00	0.00	47.90	332.00								4,050.00	3,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal OID

Fresh Water

Sample description: Fresh Water

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	0.50	0.50	0.50								1.00	10

IW 1

IW 1

Fresh Water

Sample description: Fresh Water

Sample date: 08/31/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								454.00	290
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

1

Soils

Sample and source description: Soils

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00				
DL			4.00				

2

Soils

Sample and source description: Soils

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00				
DL			4.00				

3

3

Soils

Sample and source description: Soils

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00				
DL			4.00				

East

Soils

Sample and source description: Soils

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

West

Soils

Sample and source description: Soils

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 10/28/2022: Oats, silage-soft dough

1 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	810.00	5,540.00		9.40
DL	100.00	100.00	100.00		0.01

2 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	810.00	5,540.00		9.40
DL	100.00	100.00	100.00		0.01

3 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	810.00	5,540.00		9.40
DL	100.00	100.00	100.00		0.01

East - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,430.00	790.00	5,680.00		8.10
DL	100.00	100.00	100.00		0.01

East - 06/14/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,090.00	1,050.00	13,300.00		13.70
DL	100.00	100.00	100.00		0.01

West - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,770.00	940.00	6,170.00		8.60
DL	100.00	100.00	100.00		0.01

West - 06/14/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,840.00	1,110.00	8,300.00		17.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

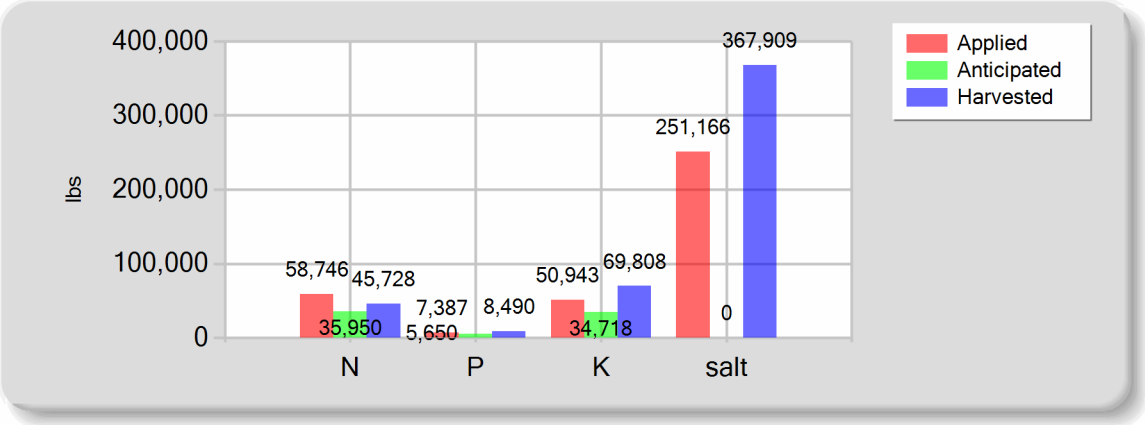
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

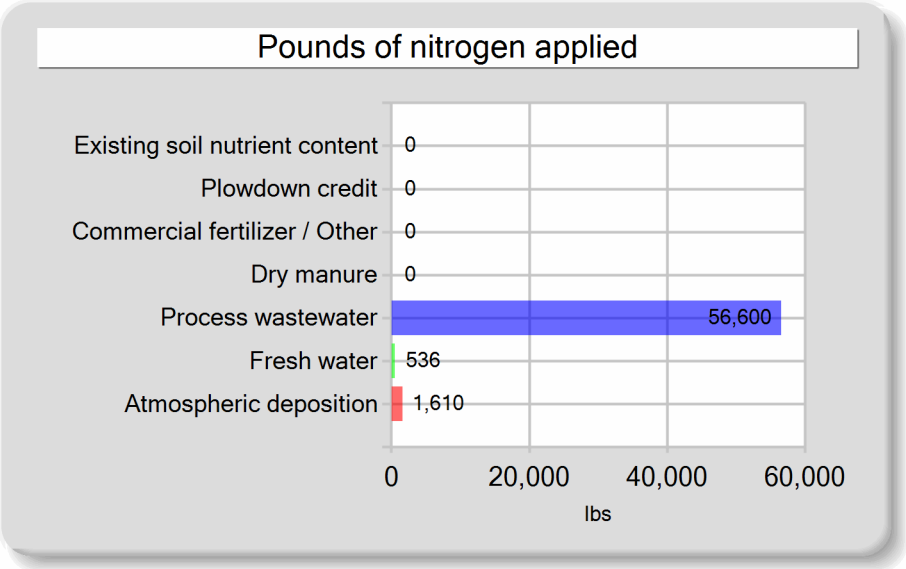
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

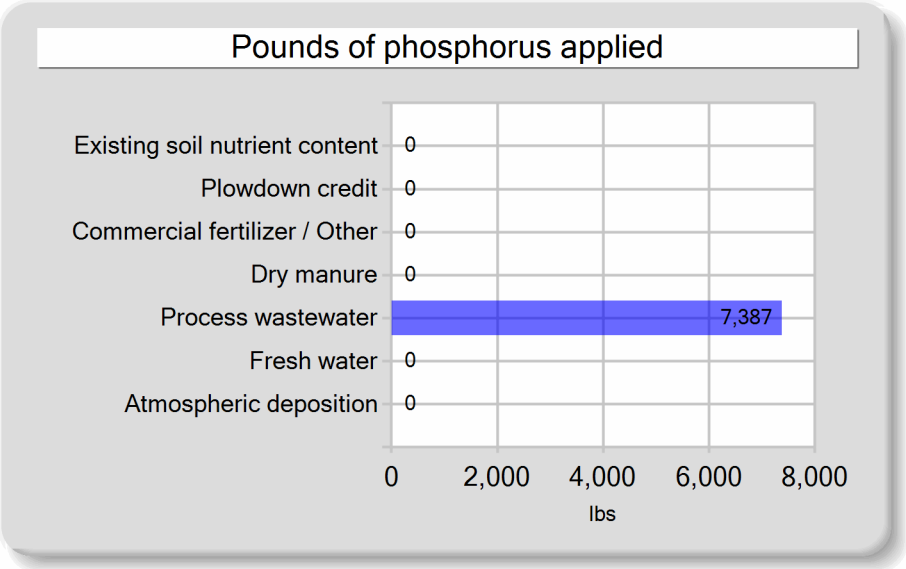
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	56,600.34	7,386.79	50,942.72	223,567.56
Fresh water	536.05	0.00	0.00	27,598.58
Atmospheric deposition	1,610.00	0.00	0.00	0.00
Total nutrients applied	58,746.39	7,386.79	50,942.72	251,166.14
Anticipated crop nutrient removal	35,950.00	5,650.00	34,717.50	0.00
Actual crop nutrient removal	45,727.99	8,490.24	69,808.47	367,909.47
Nutrient balance	13,018.40	-1,103.45	-18,865.75	-116,743.33
Applied to removed ratio	1.28	0.87	0.73	0.68

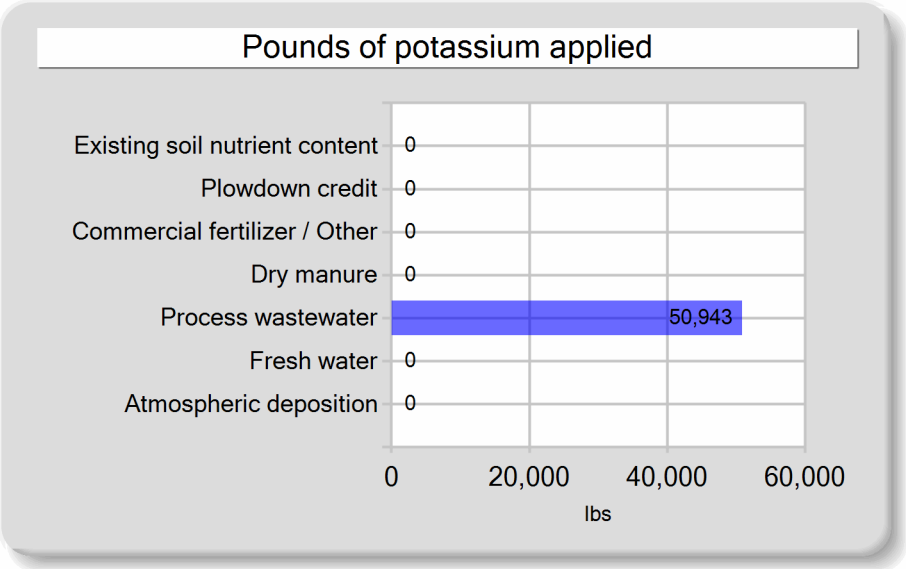
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

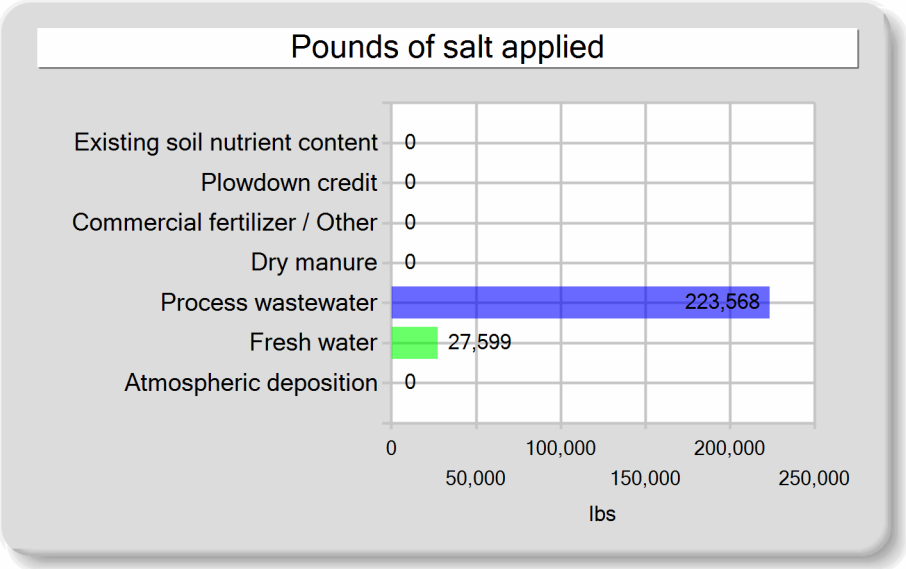


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

R. L. Sikkema	Tom Koolhaas Tom Koolhaas (Jun 13, 2024 14:13 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ralph Sikkema	Tom Koolhaas
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-13-24	Jun 13, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Tom Koolhaas

Name of Dairy Facility: River Oak Dairy #2

Facility Address: 9161 Warnerville Road, Oakdale, CA 95361
Number and Street City Zip Code

Contact Person Name and Phone Number: Tom Koolhaas 209-324-1241
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Caton Ag, Inc.

Address of Hauling Company /Person: 14303 S. Campbell Road, Escalon, CA 95230
Number and Street City Zip Code

Contact Person: Matt Caton 209-495-7769
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Advanced Soils 14303 S. Campbell Road, Escalon, CA 95320 209-499-3639
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

14303 S. Campbell Road, Escalon, CA 95230 207-320-20
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 8/15/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1777.5 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 56.9% moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

Haulers Calculations

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Tom Koolhaas Date: Jun 13, 2024

Hauler's Signature: Matt Caton Date: 8/31/2023

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Tom Koolhaas

Name of Dairy Facility: River Oak Dairy #2

Facility Address: 9161 Warnerville Road, Oakdale, CA 95361
Number and Street City Zip Code

Contact Person Name and Phone Number: Tom Koolhaas 209-324-1241
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Caton Ag, Inc.

Address of Hauling Company /Person: 14303 S. Campbell Road, Escalon, CA 95230
Number and Street City Zip Code

Contact Person: Matt Caton 209-495-7769
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Caton Ag, Inc. 14303 S. Campbell Road, Escalon, CA 95320 209-495-7769
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

14303 S. Campbell Road, Escalon, CA 95230 207-320-20
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/2/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2615 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 41.5% moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

Haulers Calculations

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Tom Koolhaas Tom Koolhaas (Jun 13, 2024 14:13 PDT) Date: Jun 13, 2024

Hauler's Signature: Matt Caton Date: 11/10/2023

DELLAVALLE™
LABORATORY INC

W & J #2
PO Box 294
Farmington, CA 95230

Account# 00-0026175
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 08/31/2023 15:45
Reported: 09/06/2023 15:37

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H2467-01	DW Dairy	Drinking Water			08/31/2023 7:00
23H2467-02	DW Trailer	Drinking Water			08/31/2023 7:00

Default Cooler Temperature on Receipt °C: 3.9

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

W & J #2
PO Box 294
Farmington, CA 95230

Account# 00-0026175
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 08/31/2023 15:45
Reported: 09/06/2023 15:37

Sample Results

Sample: DW Dairy
23H2467-01 (Water)

Sampled: 8/31/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.49**	mmhos/cm	0.01	1		09/01/23 10:21	SM 2510 B		BEH1492
Electrical Conductivity umhos	**493**	umhos/cm	10.0	1		09/01/23 10:21	SM 2510 B		BEH1492
Ammonia (as N)	*	mg/L	0.00	1		08/31/23 07:00	Field		BEI0110
Nitrate Nitrogen as NO3N	**5.5**	mg/L	0.1	1	10	09/01/23 01:23	EPA 300.0		BEH1477
pH	**8.0**	units	1.0	1		09/01/23 10:21	SM 4500-H+	H	BEH1492
Temperature	**25.0**	°C	0.0	1		09/01/23 10:21	SM 2510 B		BEH1492
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		09/05/23 16:06	SM 4500-NH3 C		BEI0014
Total Nitrogen	**5.49**	mg/L	1.00	1		09/05/23 16:06	SM 4500-NH3 C		BEI0014

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

W & J #2
PO Box 294
Farmington, CA 95230

Account# 00-0026175
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 08/31/2023 15:45
Reported: 09/06/2023 15:37

Sample Results
(Continued)

Sample: DW Trailer
23H2467-02 (Water)

Sampled: 8/31/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.44**	mmhos/cm	0.01	1		09/01/23 10:22	SM 2510 B		BEH1492
Electrical Conductivity umhos	**439**	umhos/cm	10.0	1		09/01/23 10:22	SM 2510 B		BEH1492
Ammonia (as N)	*	mg/L	0.00	1		08/31/23 07:00	Field		BEI0110
Nitrate Nitrogen as NO3N	**4.7**	mg/L	0.1	1	10	09/01/23 01:43	EPA 300.0		BEH1477
pH	**7.8**	units	1.0	1		09/01/23 10:22	SM 4500-H+	H	BEH1492
Temperature	**25.0**	°C	0.0	1		09/01/23 10:22	SM 2510 B		BEH1492
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		09/05/23 16:08	SM 4500-NH3 C		BEI0014
Total Nitrogen	**4.70**	mg/L	1.00	1		09/05/23 16:08	SM 4500-NH3 C		BEI0014

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08/31/23 15:45 23H2467

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No | Bill To: Acct # 13 | Cons # 8/31/23

No. Samples: 2 | No of Bottles: 4

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: W&J # 2

Address: PO Box 294

City: Farmington **State:** CA **Zip:** 95230

Telephone: 209 324-1241 **Fax:**

Cell/Email:

COPY TO: F&R Ag Services
&JSawyer@Frogservices.com

REQUESTED BY: Jeanette Sawyer

PROJECT:

CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

~~DGW1~~ 8-31-23
DWW1 ++0

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	~~IW 1~~ 8-31-23	8/31/23	Start 7AM	–	
2	DW Dairy	↓		-3.9	
3	DW Trailers	↓		-2.4	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
- Correction Factor: 0°C
Calibration Due: 9/26/2023
- Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	8-31-23 1:18	8-31-23 3:42
Third				
Fourth			8/31 1545	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

W & J #2
PO Box 294
Farmington, CA 95230

Account# 00-0026175
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 08/31/2023 15:45
Reported: 09/06/2023 11:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H2469-01	IW 1	Canal Water			08/31/2023 7:00

Default Cooler Temperature on Receipt °C: -1.8
Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

W & J #2
PO Box 294
Farmington, CA 95230

Account# 00-0026175
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 08/31/2023 15:45
Reported: 09/06/2023 11:28

Sample Results

Sample: IW 1
23H2469-01 (Water)

Sampled: 8/31/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.45**	mmhos/cm	0.01	1		09/01/23 10:24	SM 2510 B		BEH1492
Electrical Conductivity umhos	**454**	umhos/cm	10.0	1		09/01/23 10:24	SM 2510 B		BEH1492
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	09/01/23 01:37	EPA 300.0		BEH1476
pH	**7.7**	units	1.0	1		09/01/23 10:24	SM 4500-H+	H	BEH1492
Total Filterable Solids (TDS)	**290**	mg/L	10.0	1		09/05/23 14:53	SM 2540 C		BEI0010
Temperature	**25.0**	°C	0.0	1		09/01/23 10:24	SM 2510 B		BEH1492
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		09/05/23 16:09	SM 4500-NH3 C		BEI0014
Total Nitrogen	ND	mg/L	1.00	1		09/05/23 16:09	SM 4500-NH3 C		BEI0014

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

08/31/23 15:45 23H2469

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No: 016175-0 Bill To: Acct # Cons # 13 59

No. Samples: 1 No of Bottles: 2

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Name: W&J# 2
Address: PO Box 294
City: Farmington State: CA Zip: 95230
Telephone: 209 324-1241 Fax:
Cell/Email:
COPY TO: F&R Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Analysis and Bottles Required: (Please indicate Analysis)
DCW1 +TD
~~DWW1~~ 8.31.23

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 1	8/31/23	Start 7 AM	1.8	
2	~~DW Dairy~~ 8.31.23	↓			
3	~~DW Trailers~~				
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	8-31-23 1:18	8-31-23 3:42
Third				
Fourth	[illegible]	DLV	8/31 1545	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

2023 Annual Report River Oak Dairy #2

Final Audit Report 2024-06-13

Created:	2024-06-13
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA61DPENf1VcsVyJpyWoD-TfFgisVC8QIg

"2023 Annual Report River Oak Dairy #2" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-06-13 - 8:44:51 PM GMT

Document emailed to riveroakdairy@gmail.com for signature
2024-06-13 - 8:46:39 PM GMT

Email viewed by riveroakdairy@gmail.com
2024-06-13 - 9:12:18 PM GMT

Signer riveroakdairy@gmail.com entered name at signing as Tom Koolhaas
2024-06-13 - 9:13:03 PM GMT

Document e-signed by Tom Koolhaas (riveroakdairy@gmail.com)
Signature Date: 2024-06-13 - 9:13:05 PM GMT - Time Source: server

Agreement completed.
2024-06-13 - 9:13:05 PM GMT

Adobe Acrobat Sign